

Complete this test on your own. You may use five pages of your own, newly made notes, a graphing calculator (not connected to the internet) and nothing else.

This test is worth 50 points with 5 points of extra credit at the end. You should read the whole thing before starting, so you can pace yourself.

Note: Several of the problems below are multi-part, with part 2 relying on part 1, etc. Where possible, I'll grade on consistency (assuming you don't have a overly-simplifying answer to part 1). If you ever don't know the answer to part 1, write down an assumption at the start of part 2, and I'll grade based on that.

1 Standing waves

3 points each = 15 points total

A cylindrical pipe of length $L = 0.85$ m is closed at one end and open at the other. It is filled with air, in which the speed of sound is 340 m/s.

Note: If you're struggling here, make some drawings.

1. In the fundamental ($n=1$) mode, the wavelength of the sound is closest to:
(A) 0.85 m (B) 1.7 m (C) 3.4 m (D) 6.8 m
2. The fundamental frequency of the pipe is closest to:
(A) 50 Hz (B) 100 Hz (C) 200 Hz (D) 400 Hz
3. The pipe also resonates at frequencies above its fundamental. The lowest of these is closest to:
(A) 150 Hz (B) 200 Hz (C) 300 Hz (D) 400 Hz
4. For the pipe's fundamental frequency to double, with the length unchanged, the speed of sound in the pipe would have to change by a factor of:
(A) $\frac{1}{2}$ (B) $\sqrt{2}$ (C) 2 (D) 4
5. Suppose the pipe were open at both ends instead, with the same length. Its new fundamental frequency would be closest to:
(A) 50 Hz (B) 100 Hz (C) 200 Hz (D) 400 Hz

2 An oscillating spring

20 points total

A block of mass $m = 2.0$ kg slides without friction on a horizontal surface, attached to a spring of force constant $k = 8.0$ N/m. Measure the block's position x from the point where the spring has its natural length, with $+x$ pointing right. At time $t = 0$ the block is found at

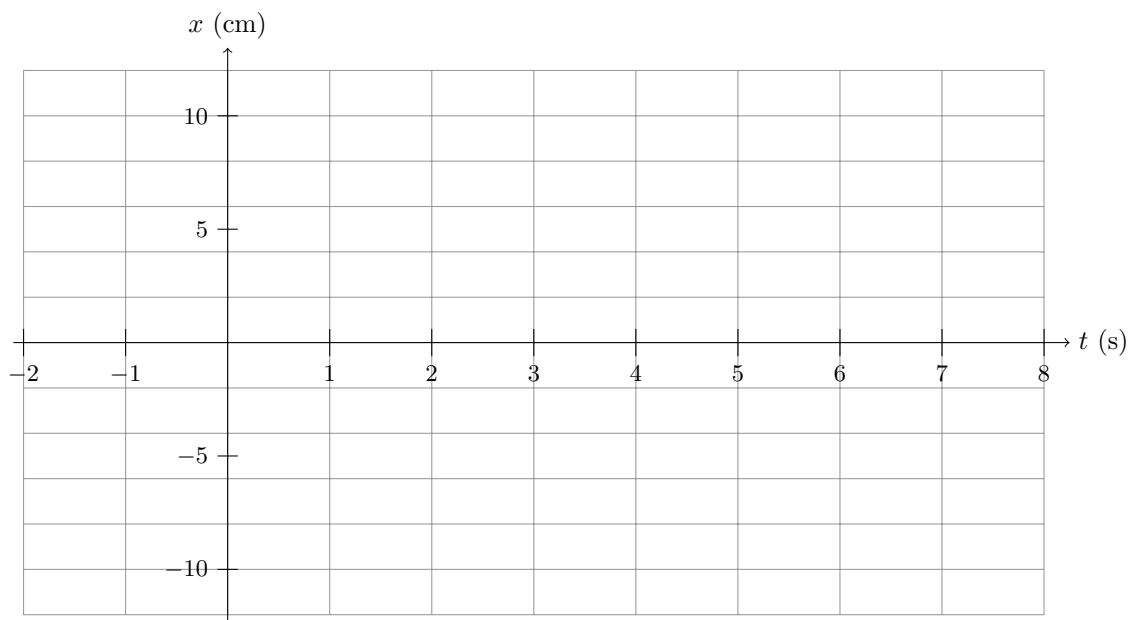
$$x_0 = +6.0 \text{ cm}, \quad v_0 = -16 \text{ cm/s}.$$

1. (3 points) Find the angular frequency ω and the period T of the motion.
2. (12 points) Find the total mechanical energy of the block in joules. Then, determine the amplitude A and the phase constant ϕ , and write the block's position as an explicit function of time,

$$x(t) = A \cos(\omega t + \phi).$$

3. (5 points) On the axes below, sketch $x(t)$ through the first two complete periods. Clearly label the amplitude, the period, and the value of x at $t = 0$.

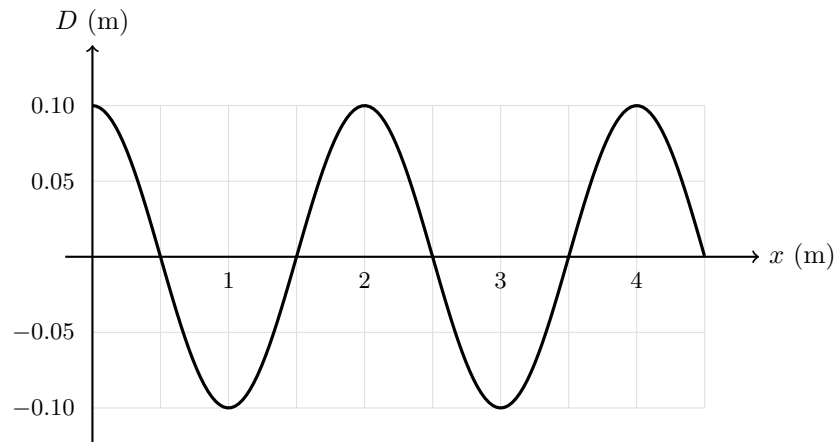
Note: I want you to plot the equation you wrote down for the previous problem. Even if that equation isn't entirely correct, if this graph is consistent with that equation, you'll receive full marks on this problem.



3 A traveling wave

15 points total

A transverse wave travels along a long string in the $+x$ direction with frequency $f = 2.0$ Hz. The figure shows the shape of the string at the instant $t = 0$; D is the transverse displacement of the string.



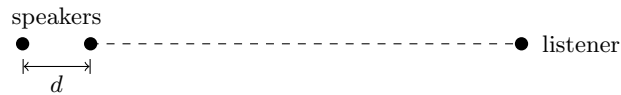
1. (3 points) Looking at the graph, what are the amplitude and the wavelength of the wave?
2. (8 points) Write the transverse displacement of the string as an explicit function of position and time, $D(x, t)$.
3. (4 points) What is the speed of the wave?

4 Extra credit - Some speakers

5 points total

Two small speakers sit side by side on a straight line. A listener stands on that same line, several meters away, beyond both speakers. Both speakers emit a steady 170 Hz tone, in phase. Take the speed of sound in air to be 340 m/s.

You now slide the speakers apart along the line by a distance d , keeping the listener on the line beyond both of them.



1. (1 points) Find the wavelength of the sound.
2. (2 points) What is the smallest separation d for which the listener hears *destructive* interference (a minimum of loudness)?
3. (2 points) What is the smallest separation d , greater than zero, for which the listener hears *constructive* interference (a maximum of loudness)?